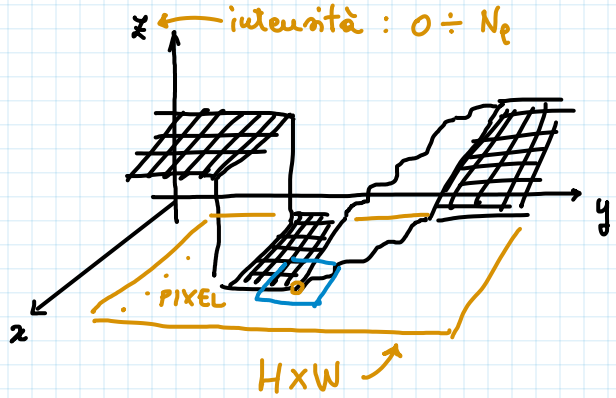
OPERATORI LOCALI

↳ FILTRI di SHARPENING ⟶ CONTORNI (EDGES) : dare NITIDEZZA

↳ FILTRI di SMOOTHING ⟶ per SFOCARE l'immagine
per Ridurre RUMORE

1) PIXEL + VICINATO

2) Riconoscere EDGES



1)

$i-1, j-1$	$i-1, j$	$i-1, j+1$
$i, j-1$	i, j	$i, j+1$
$i+1, j-1$	$i+1, j$	$i+1, j+1$

CENTRO ⟶ Pixel

Vicinato : Sotto-matrice di $L \in \mathbb{R}^{H \times W}$ quadrata e di dim n dispari.

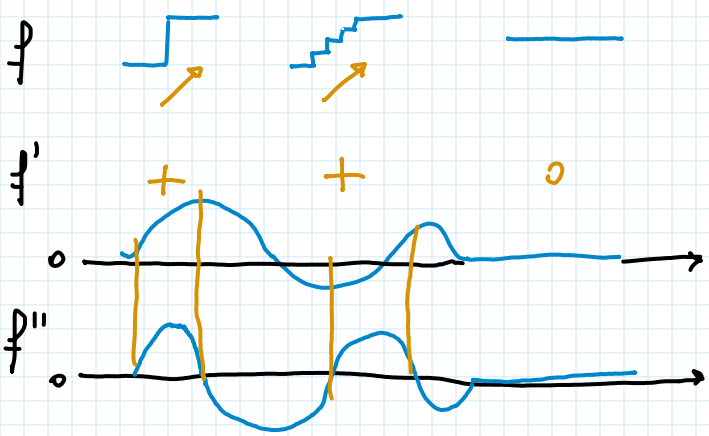
↓

Viene MODIFICATO da una MASCHERA o KERNEL o FILTRO → computer grafica

$M \in \mathbb{R}^{m \times m}$ con m dispari

Reti convoluzionali

METODO · Differenziazione spaziale

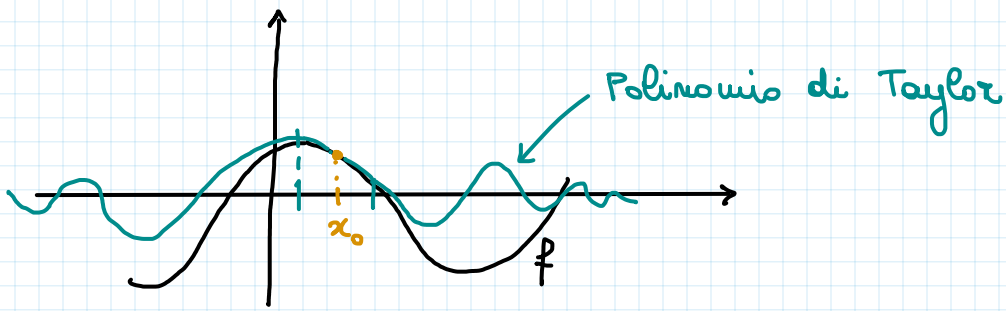


Derivata prima : RAMPE

Derivata seconda : "ATTACCO" delle RAMPE

PROBLEMA : Cosa significa calcolare ∇I dove I è una IMMAGINE DIGITALIZZATA?

SOLUZIONE : Approssimazione con POLINOMI di TAYLOR di $f : \mathbb{R}^N \to \mathbb{R}$



TEOREMA (SVILUPPO in SERIE di TAYLOR) $N=1$

Se $f: \mathbb{R} \to \mathbb{R}$ di ordine $\mathcal{C}^k(\mathbb{R})$, sia $x_0 \in \mathbb{R}$ e $x \in B(x_0, r)$ $\quad r>0 \quad r \in \mathbb{R}$

Allora:

$$f(x) = \boxed{\sum_{i=0}^{k} \frac{f^{(i)}(x_0)}{i!} \cdot (x-x_0)^i} + \overbrace{R_k(x) \cdot (x-x_0)^k}^{\text{RESTO}}$$

$$\lim_{x \to x_0} R_k(x) = 0$$

Polinomio di Taylor

ESEMPIO $N=2$

$f: \mathbb{R}^2 \to \mathbb{R} \qquad f(x,y) = x^2 + ye^x$

$$\frac{\partial^1 f}{\partial x^1} = 2x + ye^x \qquad (1,0) \qquad\qquad \frac{\partial^1 f}{\partial y^1} = e^x \qquad (0,1)$$

$$\frac{\partial^2 f}{\partial x^2} = 2 + ye^x \qquad (2,0) \qquad\qquad \frac{\partial^2 f}{\partial y^2} = 0 \qquad (0,2)$$

$$\frac{\partial^{1+1} f}{\partial x^1 \partial y^1} = e^x \qquad (1,1)$$

MULTI-INDICI

Def: Sia $f: \mathbb{R}^N \to \mathbb{R}$

$$\partial^\alpha f = \frac{\partial^{\alpha_1+\alpha_2+\dots+\alpha_N} f}{\partial x_1^{\alpha_1} \partial x_2^{\alpha_2} \partial x_3^{\alpha_3} \cdot \partial x_N^{\alpha_N}}$$

α MULTI-INDICE

$= (\alpha_1, \alpha_2, \dots, \alpha_N)$

$|\alpha| = \alpha_1 + \alpha_2 + \dots + \alpha_N$

TEOREMA di TAYLOR

Se $f: \mathbb{R}^N \to \mathbb{R}$, $f \in \mathcal{C}^k(A)$ con $A \subseteq \mathbb{R}^N$ aperto, dato $\underline{x}_0 \in A$

Per $\underline{x} \in B(\underline{x}_0, r)$ $\quad r > 0$, $r \in \mathbb{R}$, si ha:

$$f(\underline{x}) = \underbrace{\boxed{\sum_{|\alpha| \le k} \frac{\partial^\alpha f(\underline{x}_0)}{\alpha!} \cdot (\underline{x} - \underline{x}_0)^\alpha}}_{\text{POLINOMIO di TAYLOR}} + \sum_{|\alpha| = k} \underbrace{R_\alpha(\underline{x})}_{\text{RESTO}} \cdot (\underline{x} - \underline{x}_0)^\alpha$$

$\alpha! = \alpha_1! \alpha_2! \dots \alpha_m!$

$\alpha = (\alpha_1, \alpha_2, \dots, \alpha_m)$

$$\lim_{\underline{x} \to \underline{x}_0} R_\alpha(\underline{x}) = 0$$

ESEMPIO: $f(x,y) = x^2 + ye^x$, $\underline{k = 2}$ ORDINE II

$\underline{x}_0 = \underline{(0,0)}$ $\quad \alpha = (\alpha_1, \alpha_2) \longrightarrow \underline{(0,0)}, \underline{(0,1)}, \underline{(1,0)}, \underbrace{(1,1), (2,0), (0,2)}$

$(1,1)$

$$f(\underbrace{x,y}_{\underline{x} - \underline{x}_0:\ (x-0,\, y-0)}) \cong f(0,0) + \frac{\partial f}{\partial x}(0,0) \cdot \frac{1}{1!0!} \cdot (x-0)^1 + \frac{\partial f}{\partial y}(0,0) \cdot \frac{1}{0!1!} \cdot (y-0)^1$$

$\frac{\partial f}{\partial x} \to 2x + ye^x$, $\frac{\partial f}{\partial y} \to e^x$

$$+ \frac{\partial^2 f}{\partial x \partial y}(0,0) \cdot \frac{1}{1!1!} \cdot (x-0)^1 (y-0)^1 + \frac{\partial^2 f}{\partial y \partial x}(0,0) \cdot \frac{1}{1!1!} \cdot xy$$

$\frac{\partial^2 f}{\partial x \partial y} \to e^x$, $\frac{\partial^2 f}{\partial y \partial x} \to e^x$

$$+ \frac{\partial^2 f}{\partial x^2}(0,0) \cdot \frac{1}{2!0!} (x-0)^{\boxed{2}} + \cancel{\frac{\partial^2 f}{\partial y^2}(0,0)} \cdot \frac{1}{0!2!} \cdot (y-0)^2$$

$\frac{\partial^2 f}{\partial x^2} \to 2 + ye^x$, $\frac{\partial^2 f}{\partial y^2} \to 0$

$$= 0 + \cancel{(2 \cdot 0 + 0 \cdot e^0) \cdot x} + e^0 y + 2e^0 \cdot x \cdot y + (2 + 0 \cdot e^0) \cdot \frac{1}{2} \cdot x^2$$

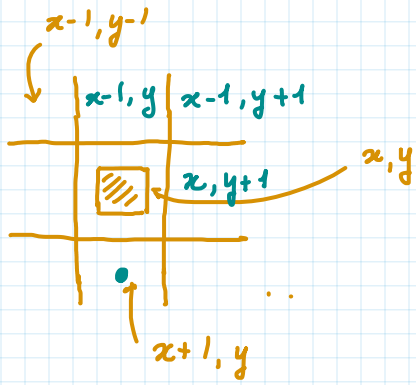
$$= y + 2xy + x^2 \qquad k = 2$$

$$\boxed{f(x,y) \cong y + 2xy + x^2}$$

ATTORNO a $(0,0)$

OPERATORI LOCALI DIFFERENZIABILI → I ORDINE : SOBEL
→ II ORDINE : LAPLACIANO

Def: FILTRO di SOBEL è un filtro del I ORDINE di dim $n \in \mathbb{N}$, $n = 3$.
↓
di SHARPENING



$\underline{x}_0 = (x, y)$

$h, k \in \mathbb{Z}$ e $\underline{x} = (x+h, y+k)$

$\underline{x}$ è in un INTORNO $B(\underline{x}_0, r)$

Polinomio di TAYLOR in $f : \mathbb{R}^2 \to \mathbb{C}$:
↓
centrato in (x, y)

$$f(x+h, y+k) \cong f(x,y) + \frac{\partial f}{\partial x}(x,y) \cdot (\cancel{x} + h - \cancel{x})^1 \cancelto{1}{(y+k-y)^0}$$
$$+ \frac{\partial f}{\partial y}(x,y) \cdot \cancelto{1}{(x+h-x)^0} (\cancel{y} + k - \cancel{y})^1$$
$$\cong f(x,y) + \frac{\partial f}{\partial x}(x,y) \cdot h + \frac{\partial f}{\partial y}(x,y) \cdot k$$

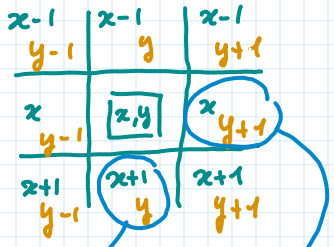
Se $k = 0$:

$$f(x+h, y+k) = f(x+h, y) \cong f(x,y) + \frac{\partial f}{\partial x}(x,y) \cdot h$$

$$\frac{\partial f}{\partial x}(x,y) \cong \frac{f(x+h,y) - f(x,y)}{h}$$

e $h = 1$:

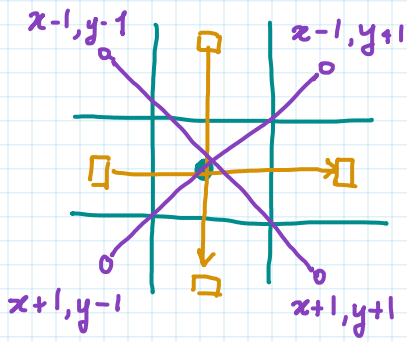
$$\frac{\partial f}{\partial x}(x,y) \cong f(x+1, y) - f(x,y)$$



Se $h = 0$ e $k = 1$:

$$f(x, y+1) \cong f(x,y) + \frac{\partial f}{\partial y}(x,y) \cdot 1$$

$$\hookrightarrow \frac{\partial f}{\partial y}(x,y) \cong f(x, y+1) - f(x,y)$$



FILTRO di SOBEL

$$M \in \mathbb{R}^{3\times 3} \quad \swarrow \quad \searrow$$

$$M_x \qquad M_y$$

$h = k = -1$

$$f(x-1, y-1) \cong f(x,y) - \frac{\partial f}{\partial x}(x,y) - \frac{\partial f}{\partial y}(x,y)$$

$h = k = 1$

$$f(x+1, y+1) \cong f(x,y) + \frac{\partial f}{\partial x}(x,y) + \frac{\partial f}{\partial y}(x,y)$$

$$\frac{\partial f}{\partial x}(x,y) + \frac{\partial f}{\partial y}(x,y) \cong \frac{f(x+1,y+1) - f(x-1,y-1)}{2}$$

HW : fare le altre APPROX.